Healthcare Management:
Patient Readmission Prediction

Analyzing 101,766 patient encounters
across 130 U.S. hospitals | 1999–2008

101,766 Patient Encounters | **50** Features | **130** Hospitals

Jack Trahey

AGENDA

01 **Introduction**

02 **Objective**
Problem statement & goals

03 **Methodology**
Data, tools & quality

04 **Analysis**
Charts, patterns & insights

05 **Findings & Conclusions**
Key takeaways

06 **Recommendations**
Action items & next steps

07 **Q&A**

Introduction

Name

Jack Trahey

Current Role

HR Professional
Cook County Health

Education

University of Illinois

Location

Chicago, IL

PROJECT CONTEXT

- Capstone project for Data Analytics program
- Business context: ReadmitGuard Analytics
- Goal: Predict 30-day patient readmission
- Tools: Excel, SQL, Python & Tableau
- Dataset: 101,766 diabetic patient encounters

Objective

🎯 GOAL

Predict the likelihood of a diabetic patient being readmitted within 30 days of hospital discharge.

⚠️ PROBLEM

Unplanned readmissions cost U.S. hospitals billions annually and signal critical gaps in post-discharge care quality.

💡 SOLUTION

Analyze 101,766 patient encounters to identify high-risk patients so providers can proactively intervene before readmission occurs.

🏥 Target Audience: Hospital administrators, clinical care teams & healthcare policy makers

Methodology

📁 DATASET

- Name: Diabetes 130-US Hospitals (1999–2008)
- Source: UCI Machine Learning Repository
- Size: 101,766 rows × 50 columns

🔧 TOOLS USED

- Excel — Statistical summary & PivotTables
- SQL — Diagnosis & readmission queries
- Python — Exploratory data analysis & viz
- Tableau — Interactive dashboard & story

🔍 DATA QUALITY FINDINGS

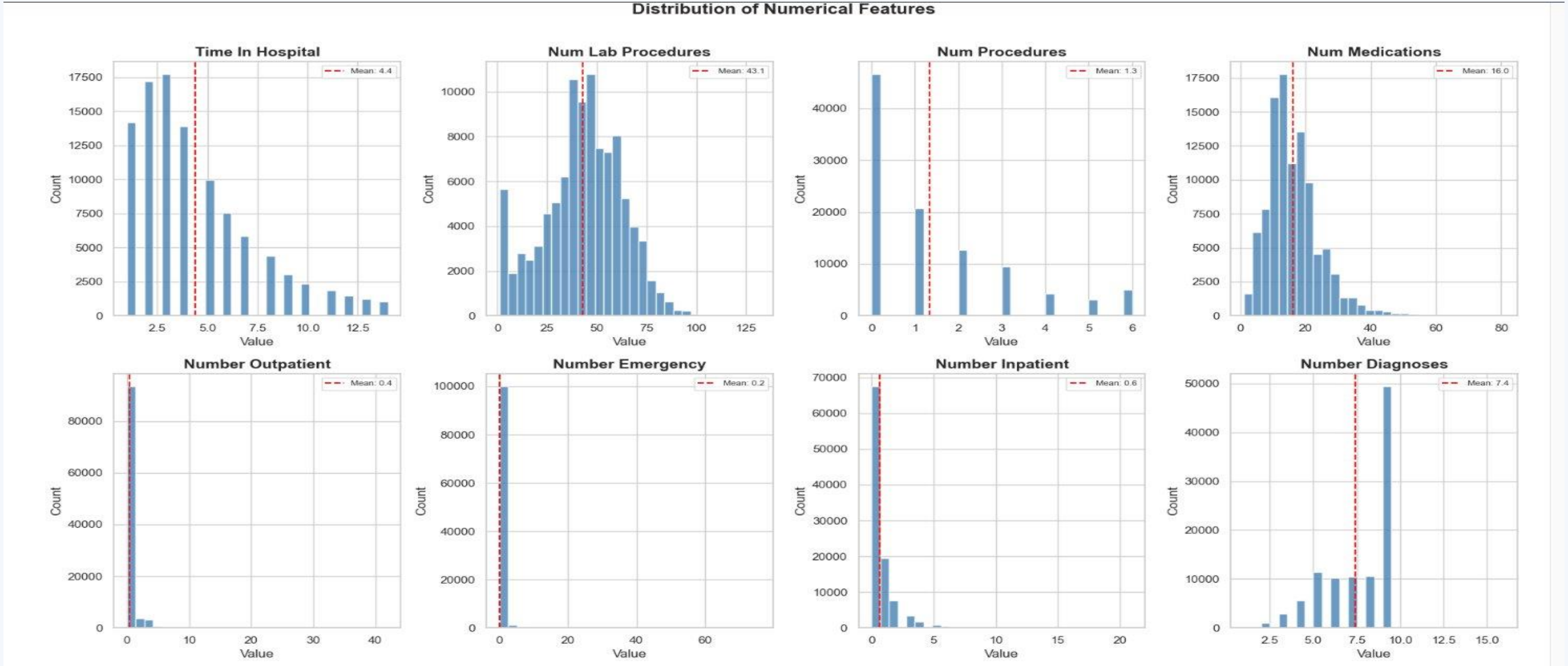
✓ Excluded 3 invalid gender records (Unknown/Invalid)

✓ Labeled 2,273 missing race values as 'Unknown/Not Reported'

✓ Excluded weight field — over 90% missing across all records

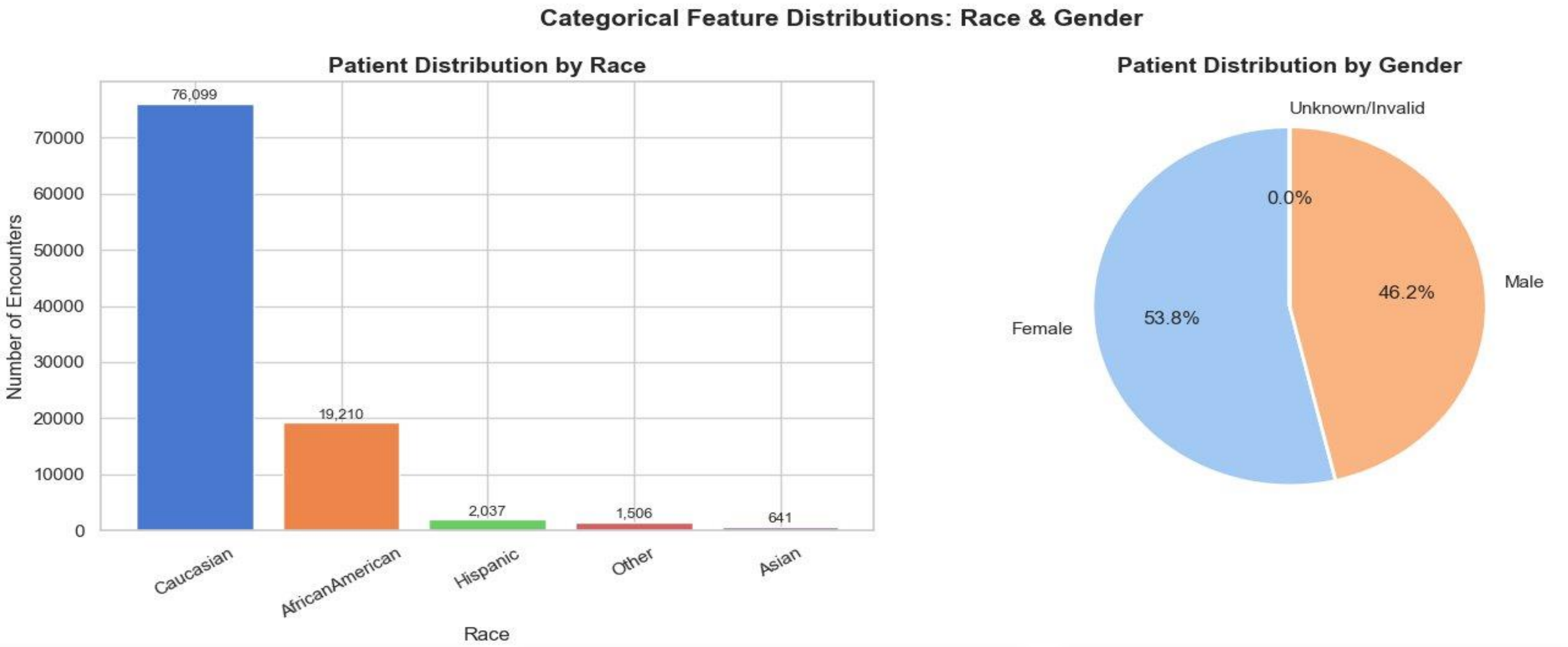
✓ Filtered '?' values from diagnosis fields in SQL queries

Analysis: Distribution of Numerical Features



Python Analysis | Key insight: Average hospital stay = 4.4 days; avg lab procedures = 43; avg medications = 16

Analysis: Race & Gender Distribution

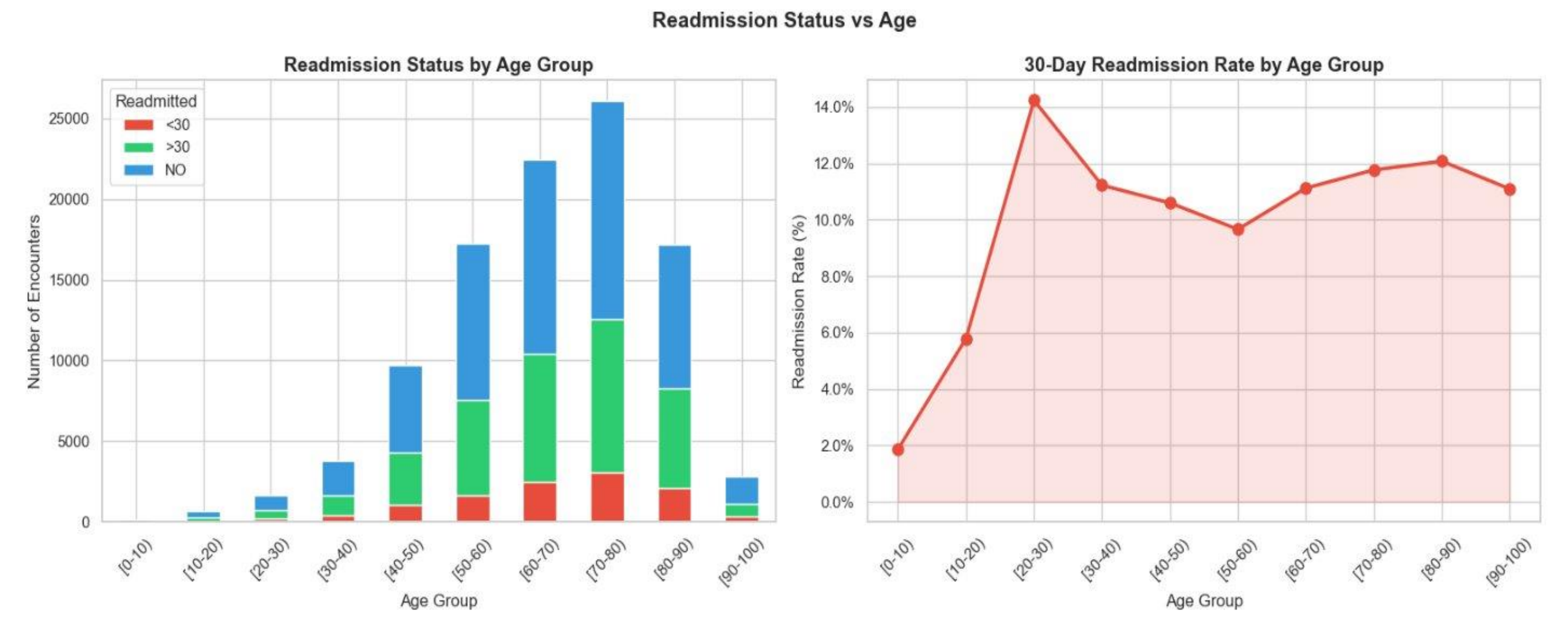


75% Caucasian patients

19% African American

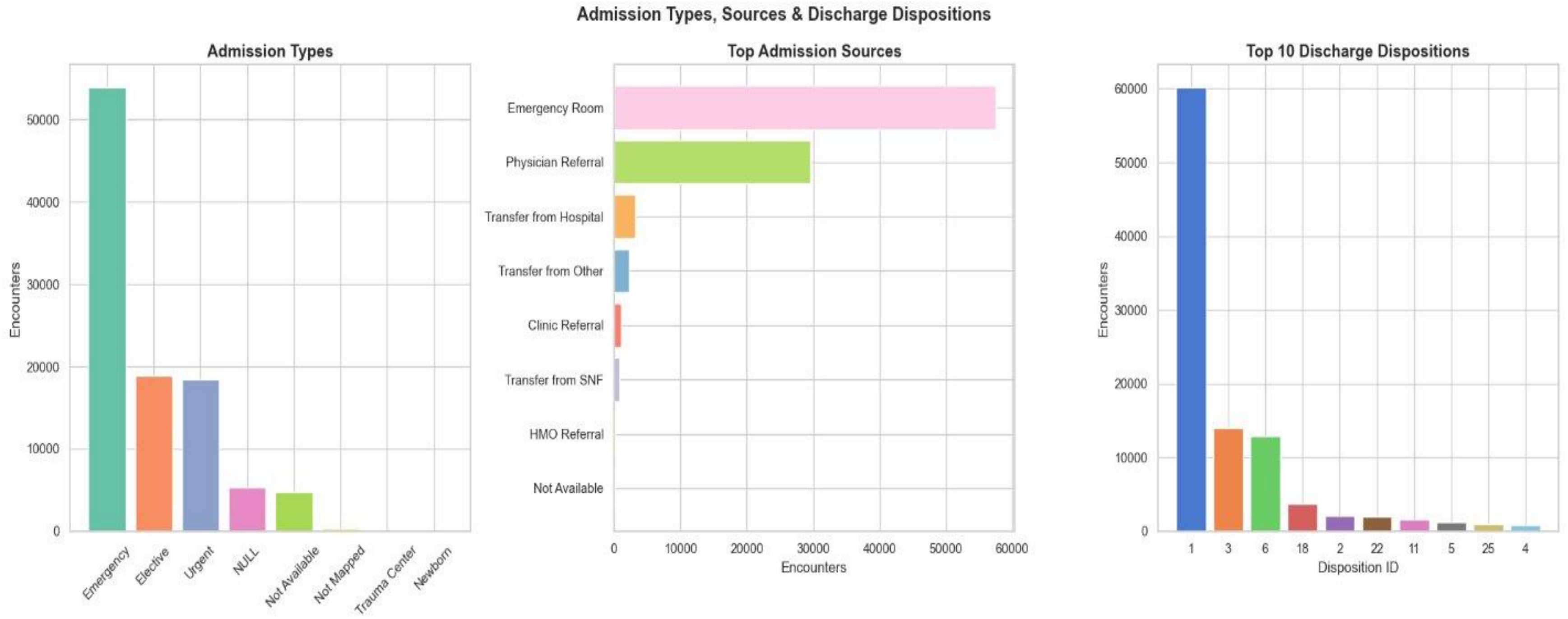
54% Female patients

Analysis: Readmission Status vs Age Group



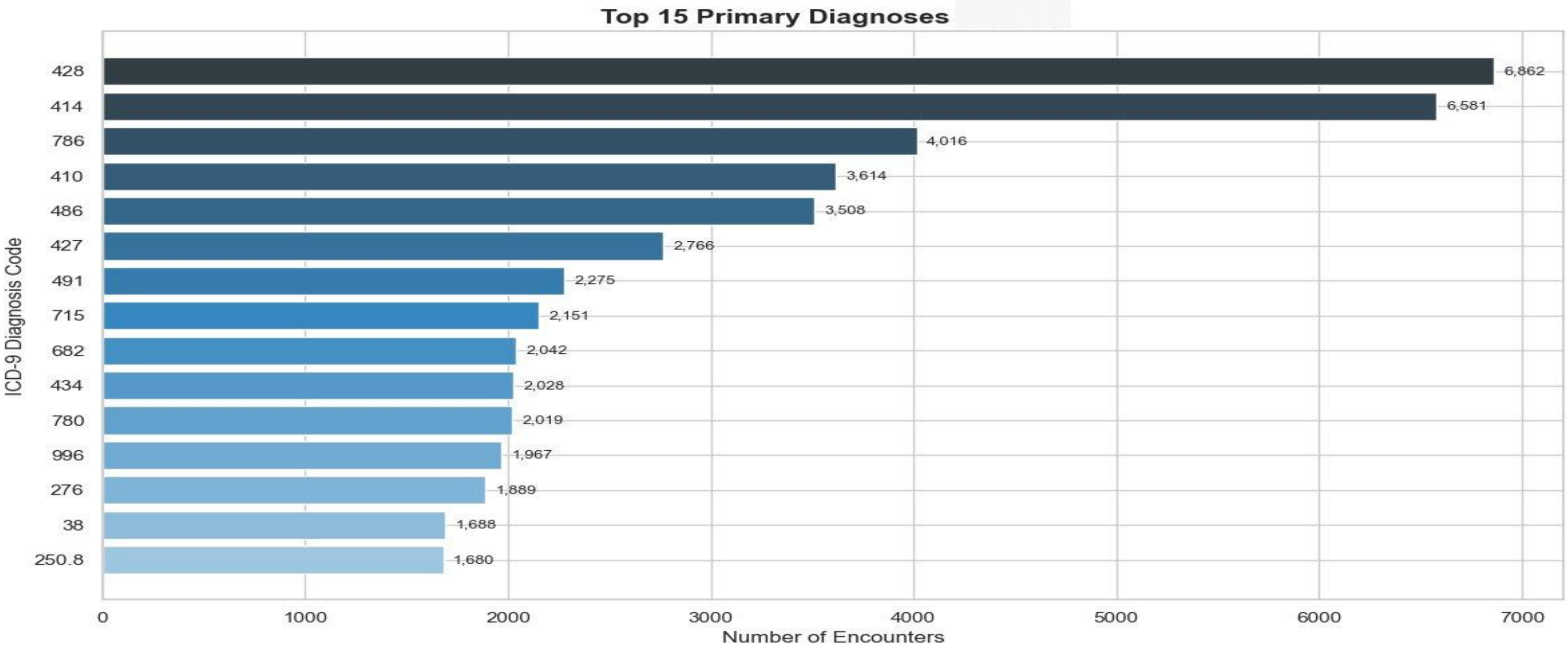
🔑 **Key Finding: Patients aged 60–80 represent the highest volume AND highest 30-day readmission rates — the primary target for intervention**

Analysis: Admission Types, Sources & Discharge



🔑 **Key Finding: Emergency admissions dominate (53,000+) with the longest average hospital stays — signaling higher acuity and readmission risk**

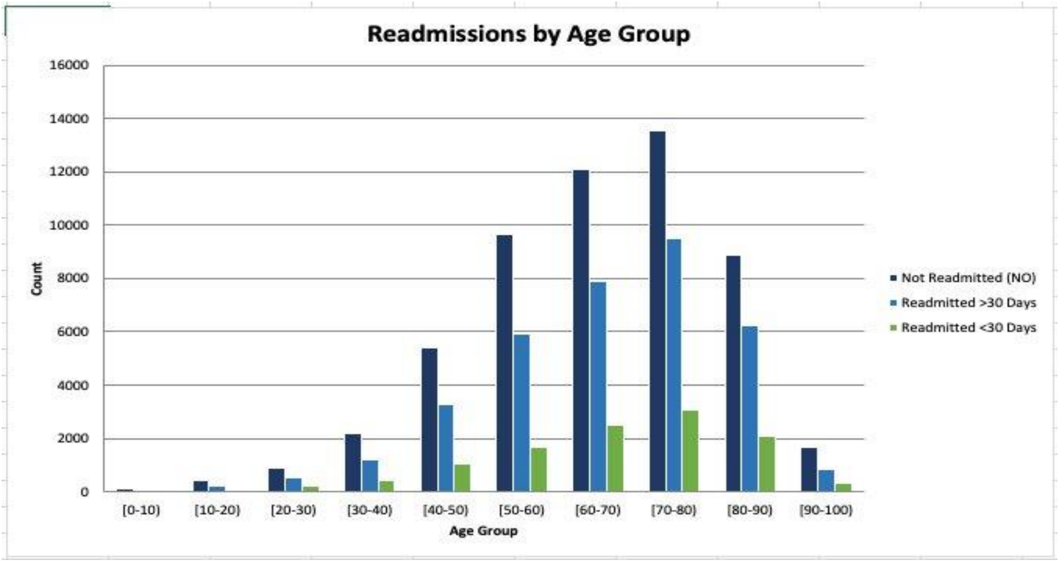
Analysis: Top Primary Diagnoses



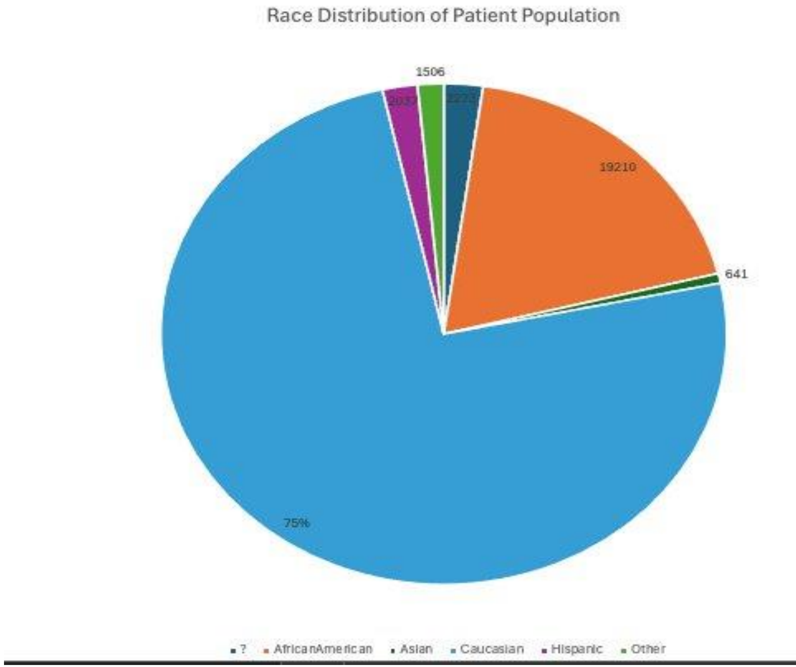
🔑 **Key Finding: ICD-9 code 428 (Heart Failure) and 414 (Coronary Artery Disease) are the top primary diagnoses — both high-readmission conditions**

Analysis: Readmissions by Age Group & Race

Readmissions by Age Group

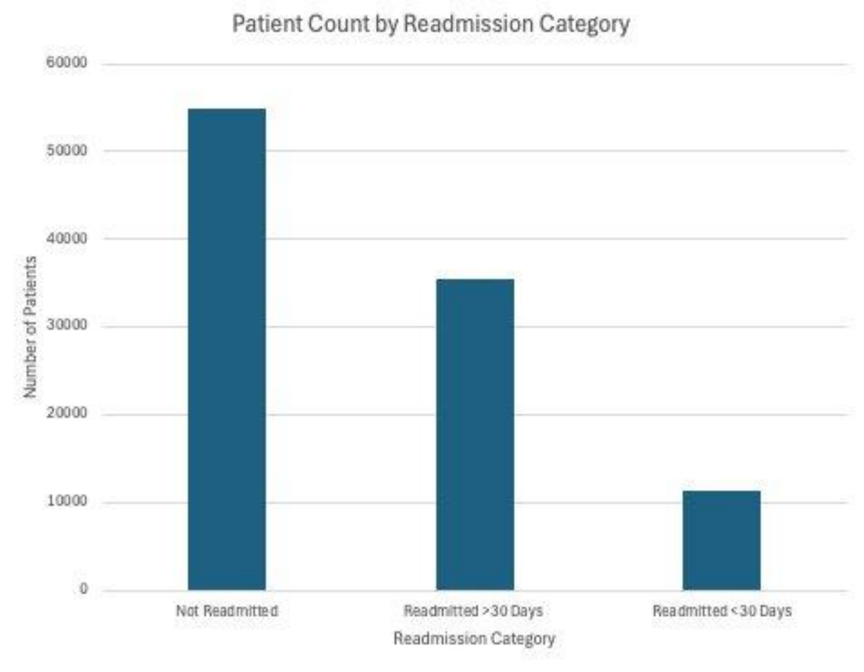


Race Distribution



Excel Analysis | Source: diabetic_data.csv

Analysis: Patient Count by Readmission Category



54% — **Not Readmitted**
54,864 patients

35% — **Readmitted >30 Days**
35,545 patients

11% — **Readmitted <30 Days**
11,357 patients

🔑 **The 11% readmitted within 30 days (11,357 patients) represent the highest clinical risk and primary focus for intervention**

Findings & Conclusions

🔴 **11%** **30-Day Readmission Rate**

11,357 patients readmitted within 30 days — the highest-risk group requiring immediate intervention

📊 **60–80** **Highest Risk Age Group**

Patients aged 60–80 have the most encounters and the highest readmission rates across all categories

💊 **16 avg** **Medications per Patient**

Higher medication counts strongly correlate with longer hospital stays and greater readmission risk

🏥 **53K+** **Emergency Admissions**

Emergency admissions dominate and carry the longest average stays, signaling higher severity and risk

💳 **Varies** **Payer Type Impact**

Readmission rates vary significantly by insurance type — Self Pay patients show elevated 30-day risk

Recommendations

01

Discharge Risk Scoring Tool

Implement a risk scoring tool at discharge that flags patients with high prior inpatient visits, multiple medications, or emergency admission types for intensive follow-up care.

02

Age-Targeted Follow-Up Protocols

Develop care transition protocols specifically for patients aged 60–80 — the highest-volume and highest-risk demographic identified in this analysis.

03

Investigate Payer Disparities

Examine payer-specific readmission rate differences to identify whether gaps in post-discharge coverage or care management programs are contributing to elevated rates.

04

Expand Data Collection

Collect social determinants of health data (housing, income, transportation) to improve future predictive model accuracy and surface root-cause risk factors.

Q&A

Questions & Answer

Thank You!

Jack Trahey